STATS C161 | Midterm Project

Trade Profitability Prediction & Trade Outlier Detection

Yuzhe He, Zhiru Guoli, Asuka Iwamoto, Lexie Zhang, Zhenlong Zhang

Public project repository: data, code, and slides

https://github.com/magnetic0flux/stats-c161-prediction-market-wallet-analysis

PRESENTATION FLOW

Agenda & Structure

01 CONTEXT

Foundation

- Core Research Problem
- Data Structure

02 ALGORITHM I

Profitability Prediction

- Predicting Profitability
- Wallet Behaviour

03 ALGORITHM II

Outlier Detection

- Isolating Anomalous Patterns
- Graphical Analysis

01 Foundation INTRO

02 Profitability ALGO I

03 Outlier Detection ALGO II

SECTION FOCUS

Research Aims

01. Predicting Profitability

Predicting **wallet profitability** based on historical betting behaviour across prediction markets.

02. Outlier Detection

Isolating **wallet entities** with anomalous or non-consensus trading activity in target markets.

29 SEP – 05 OCT 2026

Data Structure

89,736

Labelled Wallet-Market Pairs

17,177	2,661	239,434
WALLETS	MARKETS	TRADES

01 Foundation INTRO

02 Profitability ALGO I

03 Outlier Detection ALGO II

The Baseline Model
The Opening Bet:

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Labelling:

Gross profit = **Sell proceeds** − **Buy spending** + **Settlement value**

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Baseline Features:

T = moment of the wallet's first trade in this market

First batch = all its trades at **T**

1 · THE BET · FIRST BATCH

Entry price	p = $ spent ÷ shares bought (also p^2, p^3)
Bet size	log(1 + $ spent)
Number of trades	log(1 + trades)
One-sidedness	$\lvert 2s - 1 \rvert$ (s = share of $ on one outcome)

2 · THE MARKET · BEFORE T

Market volume	log(1 + $ traded before **T**)
Market traders	log(1 + wallets before **T**)
Market age	log(1 + hours open at **T**)

3 · TIMING · AT T

Hour of day	sin, cos of the hour (24-hour cycle)
Day of week	days since 29 Sep

The Baseline Model
The Opening Bet:

Labelling:

Gross profit = **Sell proceeds** − **Buy spending** + **Settlement value**

Baseline Model:

P(profitable) = logistic (**Entry price, p^2, p^3** + **Bet size** + **Number of trades** + **One-sidedness** + **Market volume** + **Market traders** + **Market age** + **Hour of day** + **Day of week**)

The bet · first batch | The market · before T | Timing · at T

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The History Model Day and Week:

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Features:

The wallet's trades in **other markets**, **before T**

Window: **24 hours** or **week-to-date** (from 29 Sep)

3 FEATURES

Prior trades	log(1 + trades)
Prior markets	log(1 + markets traded)
Prior $ bought	log(1 + $ spent buying)

+ 2 FEATURES

Prior concentration	Σ (share of $ bought in each market)^2
Prior avg buy size	log(1 + $ spent buying ÷ buy trades)

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3 FEATURES

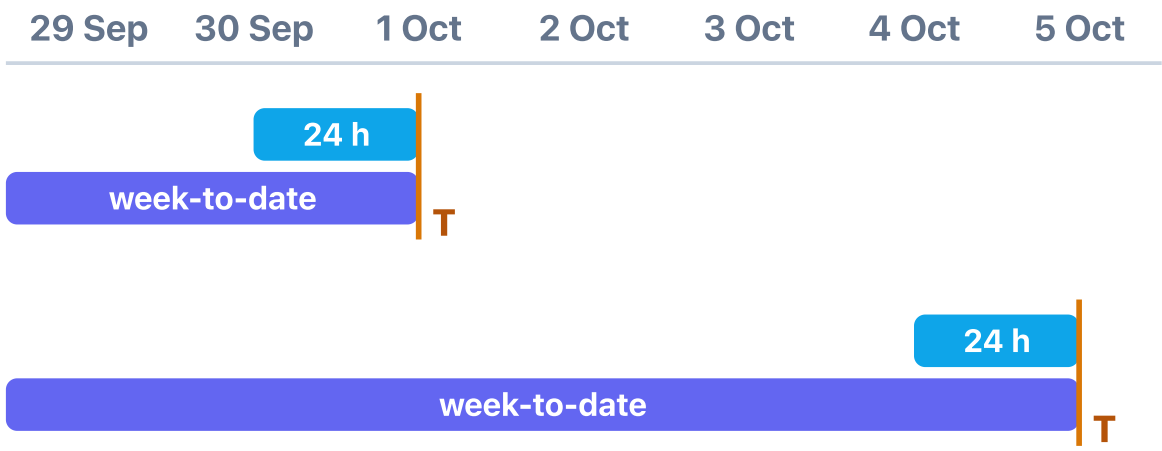
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Early bet

Late bet



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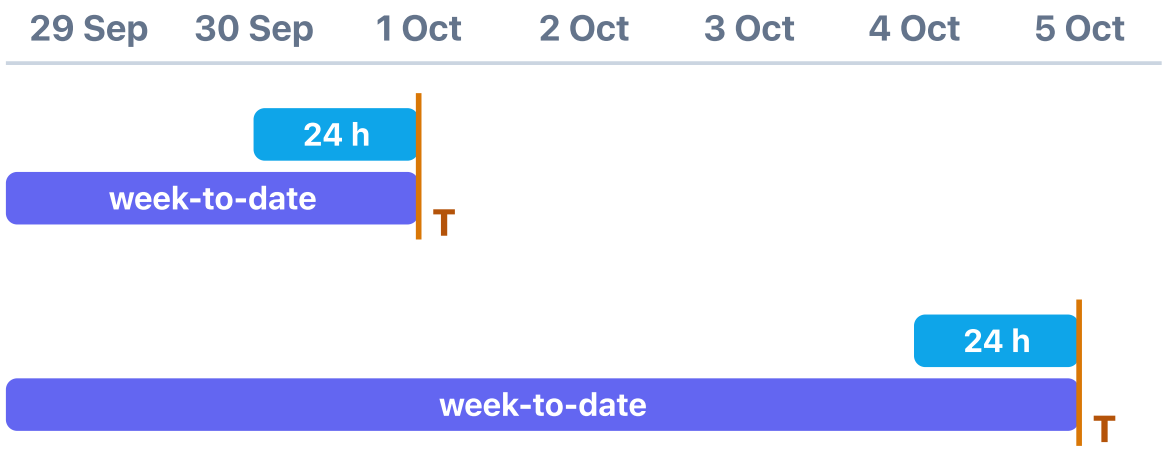
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24-hour history

Model 1	Baseline	+	3 features		
Model 2	Baseline	+	3 features	+	2 features

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Features:

The wallet's trades in **other markets**, **before T**

Window: **24 hours** or **week-to-date** (from 29 Sep)

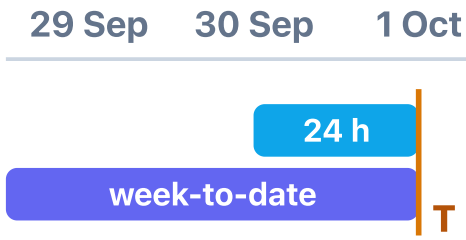
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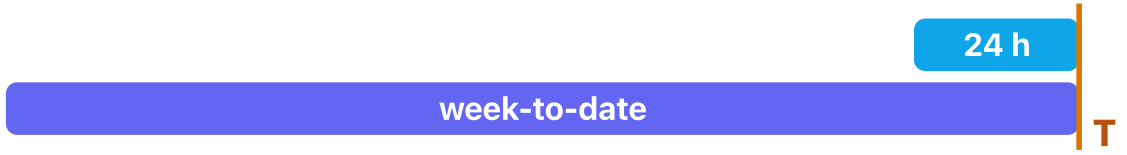
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Prior concentration	Σ (share of $ bought in each market)²
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Early bet



Late bet



24-hour history

Model 1 — **Baseline** + **3 features**

Model 2 — **Baseline** + **3 features** + **2 features**

Week-to-date history

Model 1 — **Baseline** + **3 features**

Model 2 — **Baseline** + **3 features** + **2 features**

Training & Validation:

1 · DATA · 29 SEP – 5 OCT 2026
89,736 wallet–market pairs · 16,511 wallets · 2,657 markets

▼

2 · SPLIT BY TIME
Earlier bets train, later bets test: the model never sees the future. Related markets stay in the same set.

29 Sep	30 Sep	1 Oct	2 Oct	3 Oct	4 Oct	5 Oct
History only	**Train** Fit the model			**Validate** Check	**Test** Held out · 26,507	

▼

3 · TRAIN LOGISTIC REGRESSION · 30 SEP – 3 OCT · 52,888 PAIRS

- **Baseline** — The bet, the market, timing
- **+ 24-hour history** — Wallet's trades in the 24 h before the bet
- **+ Week-to-date history** — Wallet's trades from 29 Sep up to the bet

▼

4 · TEST · 4 – 5 OCT
Compare all models on the same held-out pairs

Result:

Result:

MODEL	FEATURES	ACCURACY	ROC-AUC
Pure guessing	Always profitable	59.11%	0.50
Baseline	Baseline	**70.11%**	**0.7754**
24-HOUR HISTORY			
Model 1	Baseline + 3	70.06%	0.7752
Model 2	Baseline + 3 + 2	70.08%	0.7753
WEEK-TO-DATE HISTORY			
Model 1	Baseline + 3	70.03%	0.7754
Model 2	Baseline + 3 + 2	70.03%	0.7755

HISTORY FEATURE	24-HOUR adjusted p	WEEK-TO-DATE adjusted p
Prior trades	1.00	1.00
Prior markets	1.00	1.00
Prior $ bought	1.00	1.00
Prior concentration	1.00	1.00
Prior avg buy size	1.00	0.88
All 3 together	1.00	1.00

Result:

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24-HOUR HISTORY			
Model 1	Baseline + 3	70.06%	0.7752
Model 2	Baseline + 3 + 2	70.08%	0.7753
WEEK-TO-DATE HISTORY			
Model 1	Baseline + 3	70.03%	0.7754
Model 2	Baseline + 3 + 2	70.03%	0.7755

- The Baseline Model is 11 percentage points more accurate than pure guessing
- No wallet history improves prediction
- All history models are within 0.1 percentage points of the baseline

Result:

- No history feature is significant
- Wallet history adds no information about profitability
- A wallet's past behaviour tells us little about its profit

HISTORY FEATURE	24-HOUR adjusted p	WEEK-TO-DATE adjusted p
Prior trades	1.00	1.00
Prior markets	1.00	1.00
Prior $ bought	1.00	1.00
Prior concentration	1.00	1.00
Prior avg buy size	1.00	0.88
All 3 together	1.00	0.85

Robustness check: Random Forest

1 · SAME DATA AS LOGISTIC
Baseline + 5 history features, same split

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2 · BOOTSTRAP
Resample the training rows 200 times

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3 · GROW 200 TREES
Each split tries a random √p features

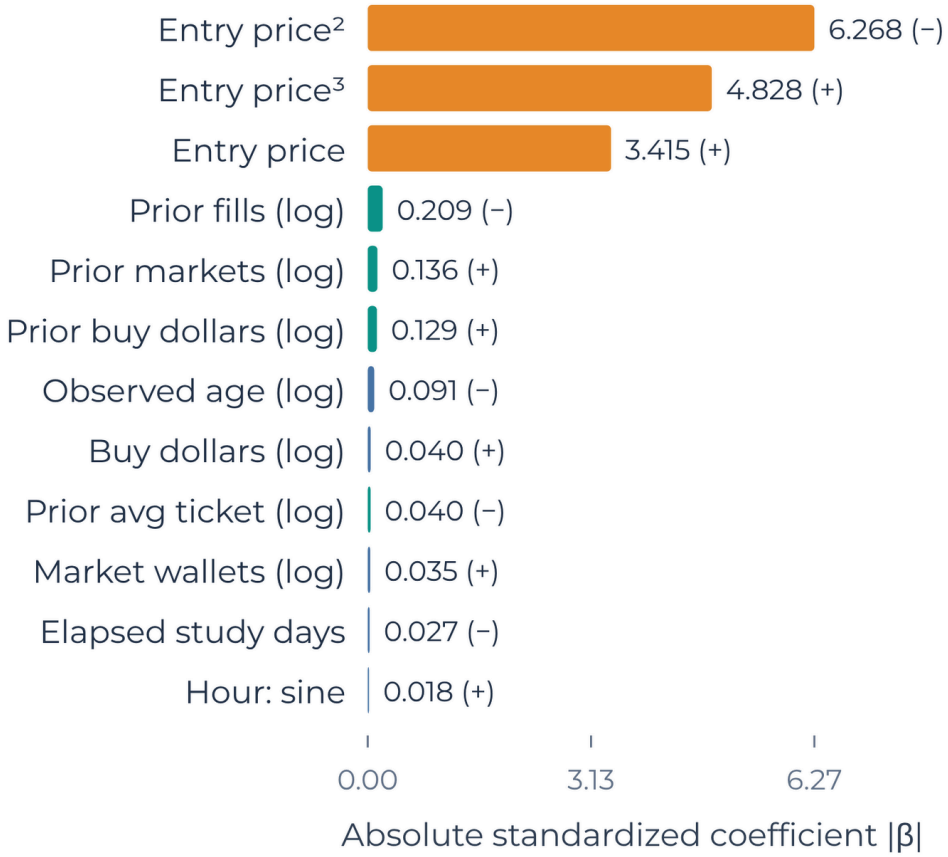
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4 · AVERAGE THE VOTES
Mean of 200 trees → P(profitable)

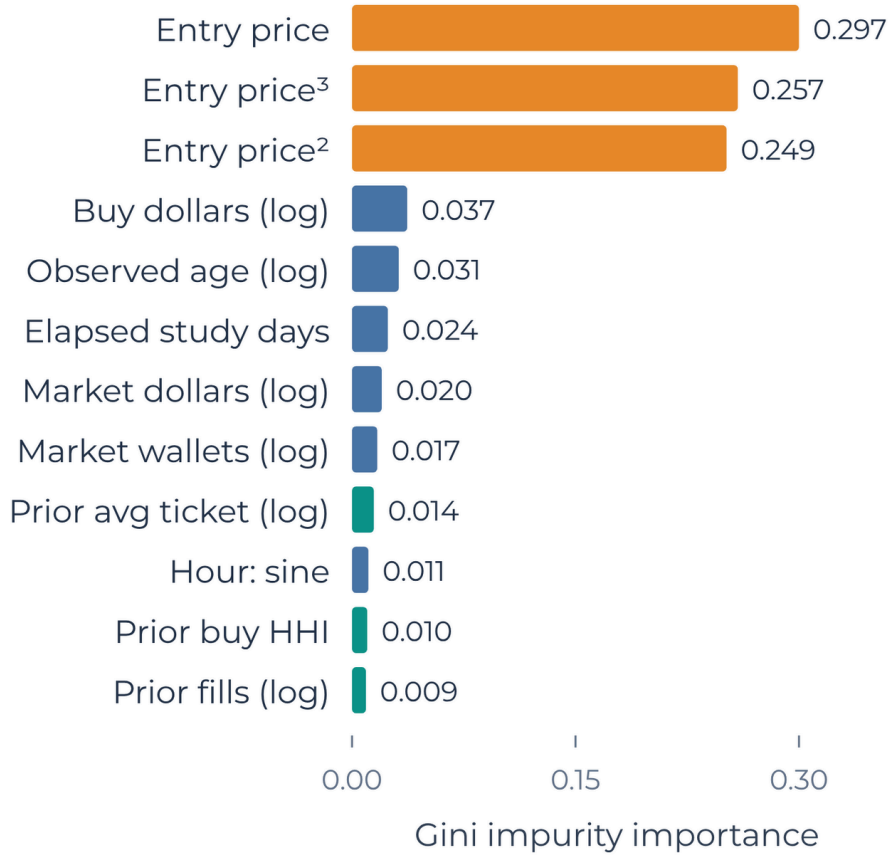
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Pure guessing	Always profitable	59.11%	0.50
LOGISTIC REGRESSION · 24-HOUR HISTORY			
Baseline	Baseline	**70.11%**	**0.7754**
Model 2	Baseline + 3 + 2	70.08%	0.7753
RANDOM FOREST · 24-HOUR HISTORY			
Baseline	Baseline	68.90%	0.7695
Model 2	Baseline + 3 + 2	70.17%	0.7730

Feature Importance: Baseline Model

Logistic regression



Random forest

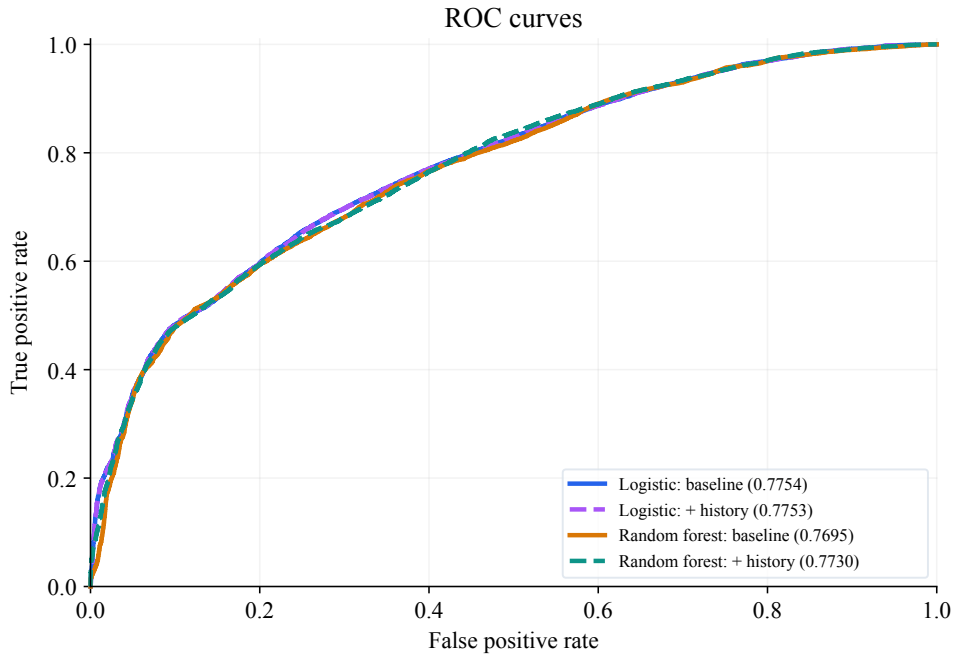


Result: Same conclusion

LOGISTIC · BASELINE
0.775

RANDOM FOREST · BEST
0.773

- Random forest does not beat logistic regression
- History only brings the forest up to the logistic baseline
- Entry price dominates both models (80% of forest importance)
- Nonlinear model finds little value in wallet history



01 Foundation INTRO

02 Profitability ALGO I

03 Outlier Detection ALGO II

1 · DATA · SAME WALLETS AS THE MODEL

16,511 wallets · 5.5 million trades

2 · 13 TRADING FEATURES PER WALLET

ACTIVITY	Volume · Trades · Markets · Active days · Active span
BET SIZE	Avg trade · Largest trade
PRICE	Avg buy price · Long-shot share · Favourite share
SELLING	Sell share
FOCUS	Market HHI · Top-market share

3 · PCA → 3D BEHAVIOUR SPACE

Standardise, keep 3 components

4 · ALPHA SHAPE → OUTLIERS

Wallets outside the shape

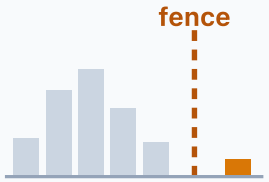
1 · MINIMUM SPANNING TREE

Link every wallet to its nearest neighbours



2 · TUKEY FENCE ON EDGE LENGTHS

Gap = edge longer than Q3 + k × IQR



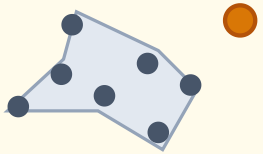
3 · CUT THE GAPS

The tree splits into clusters



4 · TOO FEW POINTS → ISOLATED

Clusters under 4 wallets are outliers



(Tukey, 1977)

AXIS	LOW SCORE = HIGH IN	HIGH SCORE = HIGH IN	READS AS
PC1 38%	Market HHI Top-market share	Markets Trades Active days	Focused ▼ Active
PC2 19%	Long-shot share	Avg trade Largest trade Volume	Small bets ▼ Big bets
PC3 17%	Long-shot share Sell share	Avg buy price Favourite share	Long shots ▼ Favourites

- Activity
- Bet size
- Price
- Selling
- Focus

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https://zhirugl.github.io/Machinelearning_Presentation/alpha_shape.html

FEATURE (median)	OUTLIERS 344	INSIDE 16,167	CLIFF'S δ	OUTLIERS ARE
PC1 · FOCUSED → SPARSE ·				
Top-market share	50%	11%	+0.61	↑ large
Markets	6	34	−0.43	↓ medium
PC2 · SMALL → BIG BETS ·				
Avg trade	$58	$8	+0.41	↑ medium
Largest trade	$278	$26	+0.42	↑ medium
PC3 · LONG SHOTS → FAVOURITES ·				
Long-shot share	6%	0.1%	+0.31	↑ small
Avg buy price	0.33	0.53	−0.28	↓ small

- Activity
- Bet size
- Price
- Selling
- Focus

(Romano et al., 2006)

Conclusion

- Outliers make a few large, concentrated bets on long shots
- Fewer outliers are profitable than other wallets (45.6% vs 57.5%).
- Outliers are a mix of potential insiders and high-risk gamblers
- Focus on outliers that win their long-shot bets

Main Takeaways

- A wallet’s opening bet, mainly its entry price, predicts its profitability.
- Wallet history provides little information about a wallet's profitability
- Trading behaviour can identify outliers
- Outliers are a combination of potential inside traders and high-risk gamblers
- Next step: focus on profitable outliers

References

- Romano, J., Kromrey, J. D., Coraggio, J., & Skowronek, J. (2006). *Appropriate statistics for ordinal level data: Should we really be using t-test and Cohen's d for evaluating group differences on the NSSE and other surveys?* Paper presented at the annual meeting of the Florida Association of Institutional Research.

- Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley.